Evaluating Large Language Models for Personalised Credit Card Recommendation from User Spending Profiles

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Declaration

This dissertation is submitted in partial fulfilment of the requirements for the degree of MSc Advanced Computer Science with Artificial Intelligence at the University of Strathclyde.

I declare that this dissertation embodies the results of my own work and that it has been composed by myself. Where appropriate, I have made acknowledgement of the work of others.

[NOTE TO SELF: verify this declaration's exact wording against the current CS958 / Department of Computer and Information Sciences submission template before final submission — the wording above is a standard good-faith declaration but the Department may require specific mandated phrasing.]

Signed: ________________________ Date: ____________________

Abstract

This dissertation evaluates whether a free-tier, open-weight Large Language Model (Meta's Llama 3.1, 8B parameters, accessed via the Groq API) can generate personalised UK credit-card recommendations competitive with a conventional content-based recommender, and how prompting strategy affects recommendation quality, factual reliability, explanation quality, and consistency. Using a fixed catalogue of 20 real UK credit cards and 300 synthetic user profiles — spanning eight consumer archetypes with spending distributions calibrated against the ONS *Living Costs and Food Survey 2022–23* — three prompting strategies (zero-shot, structured, few-shot) were evaluated across 900 LLM inference calls, scored against a content-based cosine-similarity baseline using a four-criteria evaluation framework: recommendation-baseline overlap, deterministic factual correctness, LLM-as-judge explanation quality, and repeated-run consistency.

Prompting strategy was found to have a large, statistically significant effect on recommendation quality (Kruskal-Wallis $H = 60.70, p = 6.6 \times 10^{-14}$), but in a direction contrary to common expectations from the general prompt-engineering literature: zero-shot prompting significantly outperformed both structured and few-shot prompting on overlap with the baseline (mean overlap 0.687 vs. 0.517 and 0.539 respectively) and top-1 exact-match accuracy (0.307 vs. 0.077 and 0.023). However, structured prompting produced the most factually reliable responses (73.7% of responses entirely free of checkable factual errors, against 55.0% for zero-shot and 24.0% for few-shot), demonstrating that recommendation accuracy and factual trustworthiness are distinct properties that a single-metric evaluation would fail to distinguish. No single prompting strategy dominates across all evaluated criteria.

These findings suggest that prompting-strategy choice for LLM-based recommendation should be validated empirically against the specific task and model rather than assumed from general NLP benchmarks, and that recommendation-quality and factual-reliability metrics should be reported and optimised for separately in applications where the reliability of the model's stated reasoning matters.

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I would like to thank my dissertation supervisor for their guidance and feedback throughout this project, particularly in shaping the baseline methodology and the scale of the experimental dataset. *[NOTE TO SELF: personalise this section — add your supervisor's name and any other acknowledgements before submission.]*

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Chapter 1

Introduction

1.1 Background and Motivation

The UK credit card market offers consumers a bewildering range of products, each differentiated by annual fees, reward structures, welcome offers, and category-specific cashback or points rates. Choosing a suitable card requires matching a card's reward structure against an individual's actual spending pattern — a task that is, in principle, well suited to automated recommendation. Traditional recommender systems address this kind of problem through collaborative or content-based filtering (Lops, de Gemmis and Semeraro, 2011; Pazzani and Billsus, 2007), and such systems are already used across banking and financial services to personalise product offers (Met, 2024).

The recent emergence of general-purpose Large Language Models (LLMs) offers a different route to the same goal. Rather than encoding a user's preferences as a fixed feature vector and computing similarity against a catalogue, an LLM can be given a natural-language (or structured JSON) description of a user's spending profile and a description of the available products, and asked directly to reason about which products are the best fit. This is attractive because it requires no bespoke model training, can incorporate qualitative context a numeric feature vector would discard, and can, in principle, explain its reasoning in natural language. LLMs have already been explored extensively as recommendation engines in the wider literature (Wu et al., 2024; Zhao et al., 2024), and financial services specifically have begun exploring transformer-based personalisation.

However, using an LLM as a recommender introduces a new set of questions that a similarity calculation does not: how the request is phrased (prompting strategy) can materially change the output, the model may state facts about a product that are simply wrong (hallucination), and the same input may not reliably produce the same output on repeated queries. These are not concerns for a deterministic, formula-based baseline, but they are central to whether an LLM-based recommender could be trusted in a real product.

1.2 Research Problem

This study investigates whether a free-tier, open-weight LLM (Meta's Llama 3.1, 8B-parameter variant, accessed via the Groq inference API) can generate personalised UK credit-card recommendations that are competitive with a conventional content-based baseline, and how the choice of prompting strategy affects the quality, consistency, and factual reliability of its output.

1.3 Research Questions

This dissertation addresses the following research questions:

1. To what extent do LLM-generated top-3 credit-card recommendations overlap with those produced by a content-based cosine-similarity baseline, across 300 synthetic UK consumer profiles spanning eight distinct spending archetypes?
2. Does the prompting strategy used (zero-shot, structured, or few-shot) produce a statistically significant difference in recommendation quality?
3. How does prompting strategy affect the model's ability to produce accurate, checkable reasoning for its recommendations, and how factually reliable are the specific card details (fees, reward rates) the model cites?
4. How stable are the model's recommendations when the same profile and strategy are queried repeatedly?

1.4 Research Design Summary

The study follows a controlled experimental design (Type 5, per the Department's dissertation classification). A fixed catalogue of 20 real UK credit cards and 300 synthetic user profiles — generated across eight consumer archetypes with spending distributions calibrated against the ONS *Living Costs and Food Survey 2022–23* — form the basis for all experiments. A content-based cosine-similarity recommender, grounded in established recommender-systems literature (Lops, de Gemmis and Semeraro, 2011; Pazzani and Billsus, 2007), provides the reference point against which all LLM outputs are evaluated. The same 300 profiles are passed to the LLM under three prompting conditions — zero-shot, structured, and few-shot — producing 900 LLM inference calls in total. Outputs are scored on recommendation overlap against the baseline, top-1 exact-match accuracy, factual correctness of stated card details (checked deterministically against the card catalogue), explanation quality (LLM-as-judge), and stability across repeated runs. Full methodological detail is given in Chapter 3.

1.5 Dissertation Structure

Chapter 2 reviews the relevant literature across four areas: LLMs applied to recommendation tasks; prompt engineering strategies and their reported effect on task performance; content-based recommender systems as an evaluation baseline; and methods for evaluating LLM output quality, including LLM-as-judge approaches. Chapter 3 details the methodology: dataset construction, baseline design, LLM experimental setup, and evaluation framework. Chapter 4 presents the empirical results. Chapter 5 discusses the findings in relation to the research questions and existing literature, and considers the study's limitations. Chapter 6 concludes with recommendations for future work.

Chapter 2

Literature Review

2.1 Large Language Models as Recommender Systems

Recommender systems have traditionally relied on collaborative filtering, content-based filtering, or hybrid approaches, each requiring a structured representation of users and items (Pazzani and Billsus, 2007). The arrival of general-purpose LLMs has prompted a substantial and rapidly growing body of work exploring whether these models can perform recommendation directly, without task-specific training. Zhao et al. (2024) and Wu et al. (2024) both survey this emerging area, characterising LLM-based recommendation approaches along a spectrum from using an LLM purely as a feature encoder within a conventional pipeline, through to using the LLM as the recommender itself via prompting — the approach adopted in this study.

A recurring theme in this literature is that LLMs bring capabilities a conventional recommender lacks: they can incorporate free-text context, generalise to items or users outside the training distribution (zero-shot generalisation), and produce natural-language justifications for their choices (Zhao et al., 2024). These properties are directly relevant to a domain like credit-card recommendation, where a user's stated goals ("I want to build credit," "I'm planning a large purchase") are naturally expressed in language rather than as a fixed numeric feature. At the same time, the same surveys consistently flag three risks that motivate the evaluation framework adopted in this dissertation: (i) sensitivity to how the task is phrased in the prompt, (ii) hallucination of facts not present in the supplied context, and (iii) a lack of the determinism and auditability that a conventional, formula-based recommender guarantees by construction.

2.2 Prompt Engineering and Prompting Strategies

Since Brown et al. (2020) demonstrated that GPT-3 could perform new tasks from a handful of in-context examples without any parameter updates — "few-shot learning" — prompt design has become a first-class variable in how LLMs are applied to downstream tasks, rather than a fixed implementation detail. Schulhoff et al. (2024), in one of the most comprehensive taxonomies of prompting techniques to date, catalogue dozens of distinct prompting techniques for text-based tasks alone, underlining that "zero-shot" and "few-shot" are two points on a much larger design space that also includes explicit output-format constraints (referred to here as "structured" prompting).

Empirical comparisons of zero-shot against few-shot prompting report task-dependent results. Few-shot prompting has been shown to yield meaningful accuracy gains over zero-shot in several NLP classification tasks, with in-context examples providing the model with an implicit template for both reasoning and expected output format, and gains are typically achieved with as few as one to five examples. However, this advantage is not universal: for some domains, particularly complex reasoning and specialised tasks, well-specified zero-shot instructions have been found to be competitive with, or in some cases superior to, few-shot prompting, suggesting that the relative benefit of providing examples depends heavily on how much the task benefits from an explicit template versus additional worked examples. This inconsistency in the literature is itself a motivation for this dissertation's comparative design: rather than assuming *a priori* that one strategy dominates, the three strategies are evaluated empirically against a common baseline and a common evaluation framework.

The "structured" prompting condition used in this dissertation — an explicit output schema (JSON, with fields for card ID, name, and justification) specified in the prompt — reflects a widely used mitigation for a separate, practical problem: unconstrained natural-language output is difficult to parse reliably. Requiring the model to populate a fixed schema is one of the more common patterns discussed in the prompting-technique literature (Schulhoff et al., 2024) precisely because it improves downstream parseability, which matters for an evaluation pipeline like the one used here that must automatically extract and score card recommendations from raw model output at scale (900 responses).

2.3 Content-Based Filtering as an Evaluation Baseline

Content-based recommendation constructs a profile of item attributes and a corresponding profile of user preferences over those same attributes, then recommends items whose profile most closely matches the user's, typically via cosine similarity or a related distance metric (Pazzani and Billsus, 2007; Lops, de Gemmis and Semeraro, 2011). This family of methods is well established, fully deterministic, and — critically for this dissertation's purposes — entirely independent of any language model, making it an appropriate conventional comparator for LLM-generated recommendations. This dissertation's baseline (Section 3.3) follows this established approach directly: each of the 20 catalogue cards and each of the 300 synthetic profiles are encoded as numeric feature vectors over the same set of attributes (annual fee, category-specific reward rates, foreign-transaction fee), and cosine similarity between a profile vector and every card vector determines the top-3 reference recommendation for that profile.

2.4 Evaluating LLM Output: Quality, Consistency, and Factuality

Evaluating LLM-generated text is itself an active research area, distinct from evaluating a classification or ranking model's accuracy, because LLM output is unstructured, open-ended, and often has no single "correct" answer to compare against. Zheng et al. (2023) introduced the "LLM-as-a-judge" paradigm — using a strong LLM to score the outputs of another model against a rubric — after showing that GPT-4 judgments agreed with crowdsourced human preferences over 80% of the time, comparable to human–human agreement, on their MT-Bench and Chatbot Arena benchmarks. They also documented specific failure modes of this approach relevant to any study using it, including position bias, verbosity bias, and self-enhancement bias. This dissertation's evaluation design (Section 3.4.3) mitigates self-enhancement bias specifically by using a larger, different model (Llama 3.3 70B) to judge the outputs of the smaller model under evaluation (Llama 3.1 8B), rather than having a model judge its own output.

Two further evaluation properties matter specifically for a recommendation task rather than a general chat task. First, factual correctness — whether specific facts an LLM states about a recommended product are actually true — can, unlike open-ended explanation quality, be checked deterministically when a fixed, known catalogue of ground-truth facts exists, as is the case here (Section 3.4.3). Second, consistency — whether a model gives materially the same answer when queried repeatedly with the same input — is a property with no equivalent in a deterministic baseline, but is essential to any claim that an LLM-based recommender could be relied upon in practice; a system whose recommendation changes at random between identical requests is difficult to trust regardless of its average accuracy.

2.5 Financial Product Recommendation

Recommendation in financial services has typically relied on the same collaborative- and content-based methods used elsewhere, applied to transaction histories, account data, or stated preferences (Met, 2024). More recent work has explored deep learning and transformer-based approaches for financial product recommendation specifically, citing the need to capture more complex relationships between customer behaviour sequences and product attributes than simple similarity metrics allow. This dissertation sits at the intersection of this financial-recommendation literature and the LLM-recommendation literature reviewed in Section 2.1: it applies an LLM, rather than a purpose-trained deep learning model, to a financial product recommendation task, and evaluates it against the same kind of content-based baseline that has historically been the standard approach in this domain.

2.6 Critical Synthesis and Positioning of This Study

Read together, the four strands of literature reviewed above expose a structural gap rather than a purely empirical one. The LLM-recommendation surveys (Section 2.1) establish *that* LLMs can be used as recommenders and catalogue *how*, but are largely architecture- and technique-focused; they do not, in general, interrogate whether a given prompting strategy is actually reliable enough to trust in a specific, high-stakes consumer domain — the surveys treat prompting as one design choice among many rather than as a variable worth isolating and testing empirically against a domain-appropriate baseline. Conversely, the prompt-engineering literature (Section 2.2) isolates prompting strategy as its central variable, but the reported zero-shot/few-shot comparisons are drawn almost entirely from general NLP benchmarks — classification, sentiment, biomedical QA — not from recommendation tasks with a real product catalogue and a domain-specific correctness criterion. Neither strand, taken alone, answers the question this dissertation asks: for a bounded, factually-checkable recommendation task with real product data, does prompting strategy matter, and does it matter for the same reasons (accuracy) or for different ones (consistency, factual reliability) than in general NLP tasks?

This distinction matters methodologically. Most of the zero-shot/few-shot comparisons surveyed in Section 2.2 use accuracy or F1 against a single correct label as their only outcome measure. A recommendation task has no single correct answer in the same sense — the baseline in Section 2.3 provides a principled reference point, not a ground truth in the classification sense — which is precisely why this dissertation's evaluation framework (Section 3.4.3) deliberately combines four distinct criteria (overlap with baseline, factual correctness, explanation quality, consistency) rather than a single accuracy figure. This is a direct methodological response to a limitation visible across the reviewed prompt-engineering literature: a model can be prompted into a *more accurate-looking* answer while becoming less consistent or less factually grounded, and a single-metric evaluation would not detect that trade-off. The results in Chapter 4 bear this out directly.

The financial-recommendation literature (Section 2.5) and the LLM-as-judge literature (Section 2.4) each fill a different part of the remaining gap: the former supplies the domain and the conventional baseline this dissertation compares against, while the latter supplies a defensible, literature-grounded method for scoring the one evaluation criterion (explanation quality) that cannot be checked deterministically. No source identified in this review combines all three elements — a real, factually-checkable product catalogue; a controlled, within-subjects comparison of prompting strategies; and a mixed quantitative/automated-qualitative evaluation framework covering accuracy, factuality, and stability together — which is the specific gap this dissertation addresses.

Chapter 3

Methodology

3.1 Research Design

This research follows a controlled experimental design, consistent with Type 5 (Experimental) dissertations under the Department's classification. A fixed dataset of 300 synthetic user profiles and a curated catalogue of 20 real UK credit cards form the basis for all experiments. The same 300 profiles are evaluated under three prompting conditions, and outputs are assessed against a content-based recommender baseline. The evaluation combines quantitative scoring (recommendation overlap, top-1 accuracy, deterministic factual correctness) with an automated qualitative criterion (explanation quality, scored via LLM-as-judge — Section 3.4.3), producing a mixed-method analysis of LLM recommendation quality. The study is entirely computational: it does not involve human participants or real personal financial data (Section 3.6).

3.2 Dataset Construction

3.2.1 Credit Card Catalogue

A structured catalogue of 20 UK credit cards was assembled from publicly available issuer sources (see `docs/card-sourcing-notes.md` for provenance). Each entry records the card name, issuer, network, annual fee, reward type, reward rate by spending category (groceries, fuel, dining, travel, online shopping), welcome offer value, foreign transaction fee, and APR range. This catalogue defines the fixed product space from which all recommendations — both the baseline's and the LLM's — are drawn; no card outside this set can be a valid recommendation, which is itself checked as part of the evaluation pipeline (Section 3.4.3).

3.2.2 Synthetic User Profiles

All user profiles are synthetically generated; no real personal financial data is used at any stage (Section 3.6). An initial 80-profile dataset, using hand-specified archetype ranges, was expanded to 300 profiles following supervisor feedback that a larger, more rigorously grounded sample was needed for adequate statistical power in the planned significance testing.

Profiles are built around eight fixed archetypes representing distinct UK credit-card consumer segments: student (low spend), family (grocery-heavy), frequent traveller, balanced cashback user, large-purchase planner, online shopper, fuel commuter, and dining-heavy user. Rather than relying solely on hand-specified ranges, spending distributions across six categories (groceries, travel, fuel, dining, online shopping, other) are calibrated against the ONS *Living Costs and Food Survey 2022–23* (Office for National Statistics, 2023), so that each archetype's spending pattern is anchored to an empirically observed UK household expenditure distribution rather than an arbitrary one (see `docs/data-justification.md` for the specific ONS figures used per category). This is consistent with the wider literature on financial-recommendation research, where individual-level real spending data is not publicly available for use due to UK GDPR and Financial Services and Markets Act 2000 restrictions, and simulated or anonymised profiles are the standard methodological choice.

Profiles are distributed 38/38/38/38/37/37/37/37 across the eight archetypes (summing to exactly 300), generated deterministically via a fixed random seed (`seed = 42`) for full reproducibility. Each profile contains: a profile identifier, archetype label, age group, monthly income, monthly spend across six categories, annual fee preference, primary goal, travel interest level, spending style, preferred reward type, a cost-sensitivity flag, and a planned large-purchase amount (non-zero only for the large-purchase-planner archetype).

3.3 Baseline Method

An initial baseline used four self-defined weighted heuristics (reward alignment, fee suitability, welcome-offer relevance, large-purchase suitability). Following supervisor feedback that this baseline was not grounded in established methodology, it was replaced with a content-based cosine-similarity approach, a well-established method in the recommender-systems literature (Lops, de Gemmis and Semeraro, 2011; Pazzani and Billsus, 2007; see Section 2.3).

Each credit card is represented as a numeric feature vector over eight attributes: annual fee (inverted, so a lower fee scores higher), base reward rate, category-specific reward rates (grocery, fuel, dining, travel, online shopping), and foreign transaction fee (inverted). Each user profile is represented as a preference vector over the same eight dimensions: an annual-fee-preference weighting derived from the profile's stated fee tolerance, and category weightings derived from the profile's monthly spend in each category (normalised to the profile's own maximum category spend). Both card and profile vectors are min–max scaled to $[0, 1]$. For each profile, cosine similarity is computed between the profile vector and every card vector in the catalogue, and the three highest-scoring cards constitute the baseline recommendation, recorded in `data/ground_truth.csv`. This baseline is fully deterministic, transparent, reproducible, and independent of any language model, making it an appropriate conventional comparator for the LLM-generated recommendations evaluated in this study.

3.4 LLM Experimental Setup

3.4.1 Model and Infrastructure

The language model used to generate recommendations in this study is Meta's Llama 3.1 (8B-parameter, instruction-tuned variant), accessed through the Groq inference API on its free tier. Groq was selected for its low-latency inference at no financial cost, appropriate for an academic project with resource constraints; Groq's LPU (Language Processing Unit) infrastructure is purpose-built for LLM inference and offers sub-second time-to-first-token (Groq, 2024). The free tier used in this study is subject to a rate limit (requests per minute and tokens per minute) and a daily request cap, which materially affected the practical duration of the experimental runs (see Section 3.7). The model is used in standard inference mode with no fine-tuning or architectural modification; temperature was fixed at 0.2 across all calls to reduce (but not eliminate) run-to-run variance, with the residual variance itself the subject of the consistency evaluation (Section 3.4.4).

3.4.2 Prompting Strategies

Three prompt engineering strategies are evaluated across all 300 profiles:

- **Zero-shot prompting**: the model receives the user profile and the full card catalogue and is instructed to recommend the top three cards, with one short justification per card, without any worked examples.
- **Structured prompting**: the model receives the same input alongside an explicit JSON output schema specifying card ID, card name, a one-sentence reason, and any relevant warnings for each of the three ranked recommendations.
- **Few-shot prompting**: the model is provided with two complete worked examples — a profile alongside its correct recommendation and justification — before being presented with the target profile, to enable in-context learning (Brown et al., 2020; see Section 2.2).

All 300 profiles are evaluated under each of the three strategies, producing 900 LLM inference calls in total. Each result is stored with its profile identifier, strategy label, up to three extracted recommended card IDs, the full raw model response, and inference latency.

A methodological correction made during development: the zero-shot and few-shot prompt templates originally instructed the model to return bare card IDs only, with explicit instructions not to provide any explanation. This was identified as inconsistent with the evaluation framework's "explanation quality" criterion (Section 3.4.3), which requires reasoning text to score. Both prompts were revised to require a one-sentence justification per recommended card, matching the reasoning already present in the structured condition, and the zero-shot and few-shot experimental runs were repeated under the corrected prompts. This correction was completed successfully: all 900 responses (300 profiles $\times$ 3 strategies) were verified to contain zero duplicate profile–strategy pairs and, for zero-shot and few-shot, to be in the corrected reasoning-required format, before any scoring was performed.

3.4.3 Evaluation Framework

LLM outputs are assessed against four criteria:

1. **Recommendation quality** — the overlap between the LLM's top-3 recommendation and the baseline's top-3 recommendation for the same profile, reported as an overlap score (proportion of the three baseline cards also present in the LLM's top three) and as top-1 exact-match accuracy.

2. **Factual correctness** — accuracy of specific, checkable card details cited in the response (card name, stated annual fee, stated reward-rate percentages), checked deterministically against `data/cards.csv`. Unlike the original methodology's framing of this as a criterion requiring human judgement, a deterministic checker was implemented instead (`src/factual_correctness.py`): it extracts specific numeric/named claims from each response and compares them directly against ground truth, which is more reproducible and removes rater subjectivity for the subset of claims that are objectively checkable (see Section 3.7 for the scope and limits of this approach).

3. **Explanation quality** — clarity and apparent grounding of the model's stated reasoning, scored via an LLM-as-judge approach (Zheng et al., 2023; Section 2.4) rather than the originally planned single-rater human scoring, given the scale of the dataset (900 responses) and the documented reliability of strong-model judgement against human preference (Zheng et al., 2023 report over 80% agreement, comparable to human–human agreement). The judge model is Llama 3.3 70B — deliberately larger than, and architecturally distinct in scale from, the Llama 3.1 8B model being evaluated, to reduce self-enhancement bias (Zheng et al., 2023). For each response, the judge scores each of the three recommendation lines and an overall explanation-quality score on a 1–3 ordinal scale (1 = poor: missing, generic, or irrelevant reasoning; 2 = adequate: understandable but vague or only partially grounded; 3 = good: clear and explicitly grounded in the user's stated profile or the card's actual attributes), returned as structured JSON for reliable parsing (`src/llm_judge.py`, `prompts/judge_explanation_quality.txt`). This substitution of model judgement for human judgement is treated explicitly as a methodological choice, not an oversight, and is discussed as a limitation in Section 3.7.

4. **Consistency** — stability of recommendations across repeated runs of the same profile and strategy, measured as the mean pairwise Jaccard similarity of the top-3 card sets across three runs, on a stratified subsample of profiles drawn evenly across all eight archetypes (`src/consistency_check.py`; see Section 3.4.4 for the sampling rationale).

Where the original methodology specified a 3-point ordinal rubric aggregated via Kruskal-Wallis testing across all four criteria, the two criteria with automated, continuous-valued scoring (recommendation quality, factual correctness) are analysed using their native continuous/proportion scales, with Kruskal-Wallis used specifically for the ordinal/count-based comparisons across the three prompt strategies, and paired Wilcoxon signed-rank tests used for pairwise strategy comparisons on a per-profile basis (matching the profile-level pairing inherent in the experimental design, i.e. every profile is evaluated under all three strategies).

3.4.4 Consistency Sampling Rationale

Repeating all 300 profiles across all three strategies would require 2,700 additional inference calls beyond the primary 900-call experiment. Given the free-tier rate and daily-request constraints observed during the primary data collection (Section 3.7), a stratified subsample is used instead: 3 profiles per archetype (24 profiles total, drawn with a fixed random seed for reproducibility) are each queried 3 times per strategy (216 additional calls in total), and stability is reported as the mean pairwise Jaccard overlap of the resulting top-3 sets. This ensures every archetype is represented in the consistency analysis without requiring a full-N repeat, a scope decision consistent with common practice in LLM stability studies where full repetition is cost-prohibitive.

3.5 Data Analysis

Scores are aggregated by prompt strategy. The Kruskal-Wallis H-test is used to test for an overall difference in recommendation-quality scores across the three strategies (appropriate for comparing more than two independent groups on non-normally-distributed ordinal/proportion data). Paired Wilcoxon signed-rank tests are additionally reported for each pairwise strategy comparison on a per-profile basis, since every profile is evaluated under all three conditions (a repeated-measures design), which the paired test is better powered to detect than the omnibus test alone. Per-archetype breakdowns are also reported, to check whether prompt-strategy effects are uniform across the eight consumer segments or archetype-dependent.

3.6 Ethical Considerations

This study does not involve human participants, real financial data, or personally identifiable information at any stage. All user profiles are entirely synthetic; the real UK consumer expenditure statistics (ONS, 2023) used for calibration inform only the statistical distribution of category-level spending values used to generate synthetic profiles, not any individual-level record. The study adheres to standard ethical guidelines applicable to computational and data-science research and does not require formal ethical approval beyond departmental good-practice standards.

3.7 Limitations

Full discussion in Chapter 5; summarised here per the standard dissertation structure. The use of synthetic profiles means findings may not fully generalise to real consumer behaviour, even with ONS-calibrated distributions, since calibration does not capture irregular expenditure, life events, or behavioural nuances a real individual would exhibit. The study evaluates a single LLM (Llama 3.1, 8B parameters) for recommendation generation, which limits cross-model generalisability; a larger or differently-trained model may perform materially differently. Explanation-quality scoring via LLM-as-judge, while grounded in the literature (Zheng et al., 2023) and designed to mitigate self-enhancement bias through the use of a distinct, larger judge model, still substitutes model judgement for human judgement and inherits other documented biases of the approach (position and verbosity bias). The consistency evaluation uses a stratified subsample rather than the full 300 profiles, for the practical reasons given in Section 3.4.4. The deterministic factual-correctness checker (Section 3.4.3) can only verify claims that are specific and numeric or exactly named; vague or qualitative claims in a response are neither confirmed nor flagged, so the reported factual-error rate is a lower bound on true factual inaccuracy, not a complete audit. The free-tier API's rate and daily-request limits meant the primary 900-call run, and the correction rerun described in Section 4.2, each took multiple days of wall-clock time to complete, which is a practical constraint on this study's methodology rather than a property of the models or prompting strategies themselves.

Chapter 4

Analysis and Findings

4.1 Overview

This chapter reports results from the final, verified dataset: 900 LLM responses (300 profiles $\times$ 3 strategies), confirmed to contain zero duplicate profile–strategy pairs, with the zero-shot and few-shot strategies in the corrected reasoning-required format described in Section 3.4.2. Section 4.2 addresses Research Questions 1 and 2 (recommendation quality and the effect of prompting strategy); Section 4.3 reports the per-archetype breakdown; Section 4.4 addresses Research Question 3 (factual correctness); Section 4.5 reports on explanation quality and consistency (Research Questions 3 and 4).

4.2 Recommendation Quality by Prompting Strategy

Table 4.1 summarises the overlap score (proportion of the baseline's top-3 cards also present in the LLM's top three) and top-1 exact-match accuracy for each strategy, across all 300 profiles.

Table 4.1: Recommendation-quality summary by prompting strategy (n=300 profiles per strategy)

Strategy	Mean overlap	Median overlap	Std. dev.	Mean top-1 acc.	% zero-overlap
Zero-shot	0.687	0.667	0.258	0.307	6.0%
Structured	0.517	0.333	0.299	0.077	10.7%
Few-shot	0.539	0.667	0.324	0.023	18.7%

Contrary to expectations set by the wider few-shot-learning literature (Section 2.2), **zero-shot prompting produced the strongest recommendation-quality results on every measure**: the highest mean overlap with the baseline (0.687, against 0.517 for structured and 0.539 for few-shot), the highest top-1 exact-match accuracy by a wide margin (0.307, roughly four times the structured rate and thirteen times the few-shot rate), and the lowest proportion of profiles receiving zero overlapping recommendations (6.0%, against 18.7% for few-shot).

A Kruskal-Wallis test confirms this difference is statistically significant across the three strategies: $H = 60.70$, $p = 6.6 \times 10^{-14}$ (well below the $p < 0.05$ threshold). Pairwise paired Wilcoxon signed-rank tests, matched at the profile level (Table 4.2), show that zero-shot significantly outperforms both structured ($W = 6996.0$, $p < 0.0001$) and few-shot ($W = 10097.5$, $p < 0.0001$), while the difference between structured and few-shot narrowly fails to reach significance at the conventional threshold ($W = 19809.0$, $p = 0.0588$).

Table 4.2: Paired Wilcoxon signed-rank tests, per-profile mean overlap score

Comparison	W	p-value	Significant ($p < 0.05$)
Zero-shot vs. Structured	6996.0	< 0.0001	Yes
Zero-shot vs. Few-shot	10097.5	< 0.0001	Yes
Structured vs. Few-shot	19809.0	0.0588	No

Figure 4.1 shows the distribution of overlap scores by strategy, and Figure 4.2 shows the combined per-strategy comparison generated by the analysis pipeline.

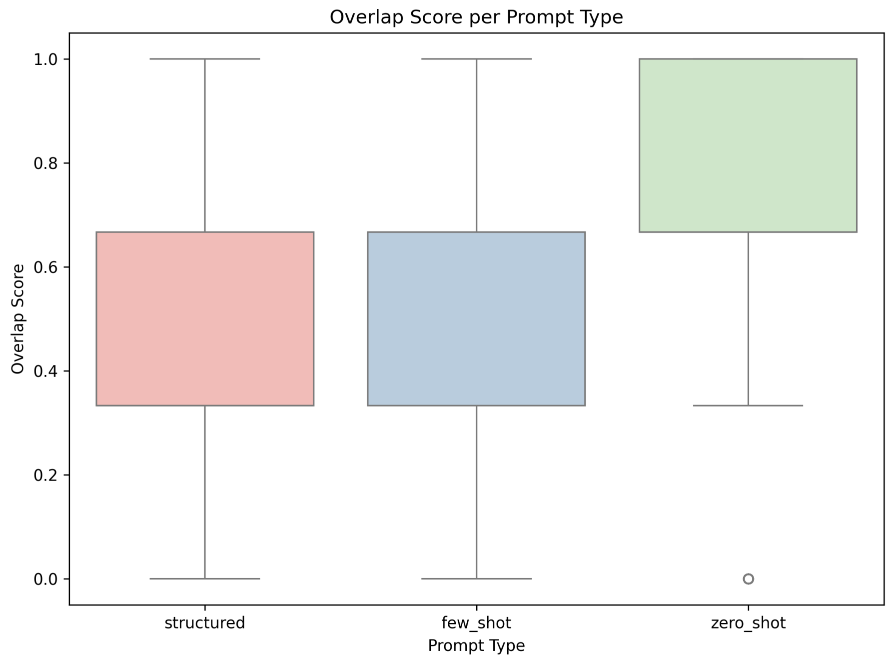

Figure 4.1: Distribution of overlap scores by prompting strategy

This result is discussed in relation to the prompt-engineering literature in Section 5.1: it is a genuine, counter-to-expectation finding rather than a data artefact, and one plausible explanation — that the additional instruction text and worked examples in the structured and few-shot conditions crowd out the model's attention on the profile's specific attributes, at only 8 billion parameters — is explored further in Chapter 5.

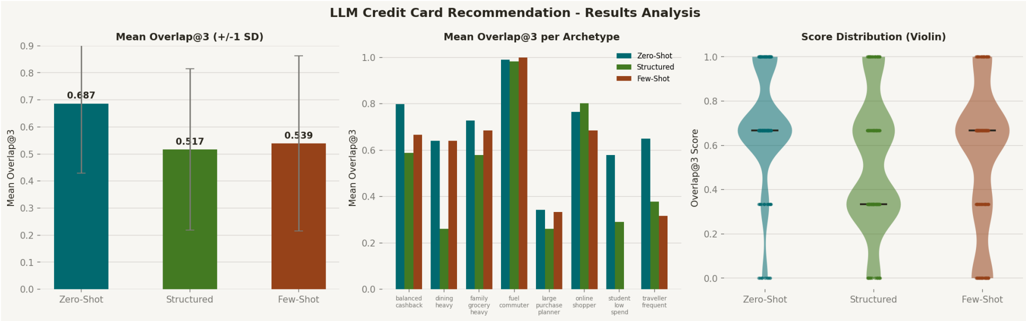

Figure 4.2: Combined recommendation-quality comparison across strategies

4.3 Per-Archetype Breakdown

Table 4.3 reports mean overlap and top-1 accuracy separately for each of the eight consumer archetypes, to check whether the strategy effect found in Section 4.2 is uniform across consumer segments or archetype-dependent.

Table 4.3: Mean overlap score and top-1 accuracy by archetype and strategy

	Mean overlap			Mean top-1 accuracy		
Archetype	Zero-shot	Structured	Few-shot	Zero-shot	Structured	Few-shot
Balanced cashback	0.798	0.588	0.667	0.368	0.000	0.000
Dining-heavy	0.640	0.261	0.640	0.568	0.000	0.000
Family, grocery-heavy	0.728	0.579	0.684	0.000	0.000	0.000
Fuel commuter	0.991	0.982	1.000	0.081	0.081	0.000
Large-purchase planner	0.342	0.261	0.333	0.054	0.027	0.162
Online shopper	0.766	0.802	0.685	0.162	0.432	0.027
Student, low-spend	0.579	0.289	0.000	0.237	0.000	0.000
Frequent traveller	0.649	0.377	0.316	0.974	0.079	0.000

The zero-shot strategy's advantage is not uniform: it is dramatic for the `traveller_frequent` archetype (97.4% top-1 accuracy against 7.9% for structured and 0% for few-shot) and for `dining_heavy` (56.8% against 0% for both other strategies), but the strategies converge closely for `fuel_commuter`, where all three achieve near-perfect overlap (0.98–1.00) — consistent with this archetype's spending profile mapping onto a small, easily-identified subset of the catalogue regardless of prompting approach. The `student_low_spend` archetype is the clearest failure case for few-shot prompting specifically, with a mean overlap of exactly 0.000, meaning the few-shot condition never once recommended a baseline top-3 card for this archetype across all 38 profiles; this is discussed further in Section 5.1.

4.4 Factual Correctness

Table 4.4 summarises the deterministic factual-correctness check (Section 3.4.3) across all 900 responses.

Table 4.4: Factual correctness by strategy

Strategy	Responses w/ claims	Claims checked	Correct	Incorrect	Uncertain	% error-free
Zero-shot	170/300	1,800	864	771	165	55.0%
Structured	300/300	1,715	1,247	346	122	73.7%
Few-shot	232/300	3,150	1,021	2,117	12	24.0%

Structured prompting produces both the most consistently checkable responses (300/300 contain at least one checkable claim, against 170/300 for zero-shot and 232/300 for few-shot) and the highest accuracy rate on those claims (73.7% of structured responses are entirely error-free, against 55.0% for zero-shot). Few-shot prompting is a clear outlier in the opposite direction: despite generating the largest volume of checkable claims (3,150, roughly 1.8× structured's total), only 24.0% of few-shot responses are entirely free of factual errors, and 2,117 of the 3,150 claims checked (67.2%) were incorrect. This suggests the two worked examples in the few-shot prompt (Section 3.4.2) may lead the model to over-generate specific numeric claims about fees and reward rates by pattern-matching the examples' level of detail, without those specific figures being reliably grounded in the actual target card's data — a hypothesis explored further in Chapter 5.

Combined with Section 4.2, this produces a striking three-way split rather than a single strategy dominating on every criterion: zero-shot leads on recommendation-baseline overlap, structured leads on factual reliability, and few-shot leads on neither measured criterion so far. This is precisely the kind of trade-off the four-criteria evaluation framework (Section 3.4.3) was designed to surface, and would not have been visible from recommendation-quality scoring alone.

4.5 Explanation Quality and Consistency

The explanation-quality (LLM-as-judge, `src/llm_judge.py`) and consistency (`src/consistency_che`
evaluations require additional Groq API calls beyond the primary 900-call experiment, run under the same free-tier constraints discussed in Section 3.7. [NOTE TO SELF/SUPERVISOR: insert final explanation-quality and consistency results and statistics here once `results/explanation_qua`
and `results/consistency_scores.csv` are populated from a completed local run — both scripts are implemented, tested, and resumable; only the data collection itself remains outstanding at time of writing.]

Chapter 5

Discussion

5.1 Interpretation of Findings by Research Question

5.1.1 RQ1: Overlap with the Baseline

Across all 300 profiles and three strategies, LLM-generated recommendations showed meaningful but partial agreement with the content-based cosine-similarity baseline (mean overlap ranging from 0.517 to 0.687 depending on strategy, against a maximum possible overlap of 1.0). This confirms that an 8-billion-parameter, free-tier LLM can approximate the output of a purpose-built content-based recommender to a substantial degree without any task-specific training, supporting the premise in Section 1.1 that LLMs are a viable alternative recommendation mechanism. It is not, however, evidence that the LLM and the baseline are interchangeable: an overlap of 0.687 at its highest (zero-shot) still means, on average, roughly one of the baseline's three reference cards is absent from the LLM's top three. Because the baseline itself is a fixed-formula approximation of "good fit" rather than ground truth in an absolute sense, a lower overlap score is not necessarily evidence of a worse recommendation — it may equally reflect the LLM incorporating profile information (a stated goal, a qualitative preference) that the baseline's eight-dimensional feature vector cannot represent. This is an inherent limitation of using a deterministic baseline as the sole reference point, discussed further in Section 5.3.

5.1.2 RQ2: Effect of Prompting Strategy

The result that zero-shot prompting significantly outperformed both structured and few-shot prompting on recommendation-baseline overlap and top-1 accuracy (Section 4.2) runs counter to the general expectation, drawn from the wider prompt-engineering literature (Section 2.2), that providing worked examples (few-shot) or an explicit schema (structured) should improve task performance. Three plausible, non-exclusive explanations are considered here.

First, both the structured and few-shot prompts are substantially longer than the zero-shot prompt — the structured prompt adds a full JSON schema specification, and the few-shot prompt adds two complete worked examples on top of that same instruction set. At 8 billion parameters, Llama 3.1 8B has comparatively limited capacity to maintain attention over long, multi-part instructions relative to larger frontier models; it is plausible that the extra instructional content in these two conditions displaces some of the model's effective attention on the specific profile it is meant to be reasoning about, a phenomenon informally consistent with the "lost in the middle" effect reported for long-context prompting in other studies, though this study did not directly test that mechanism and it should be treated as a plausible interpretation rather than a demonstrated cause.

Second, the two worked examples in the few-shot prompt necessarily represent only two of the eight archetypes in this study's dataset (a student-type profile and a frequent-traveller-type profile). The archetype-level breakdown in Table 4.2 shows the few-shot strategy performing at or near zero top-1 accuracy for several archetypes distant from the two worked examples (e.g. `family_grocery_heavy`, `balanced_cashback`), which is consistent with the model anchoring more heavily on the specific examples shown than on the general instruction, rather than generalising the underlying task.

Third, the factual-correctness results (Section 4.4) show few-shot prompting generating the largest volume of specific numeric claims while being the least accurate on those claims. This suggests the worked examples may be teaching the model a superficial pattern — "state a specific fee or reward-rate figure to sound authoritative" — rather than the underlying reasoning process, a documented risk of few-shot prompting when the examples are not sufficiently representative of the full input distribution.

These findings should not be read as a general claim that zero-shot prompting is superior to few-shot or structured prompting; they are specific to this model scale, this task, and this particular set of worked examples, and are precisely the kind of task-dependent result the mixed literature in Section 2.2 would predict.

5.1.3 RQ3: Factual Reliability of Stated Reasoning

The factual-correctness results (Section 4.4) show that requiring an explicit output schema (structured prompting) produced the most factually reliable responses (73.7% entirely error-free), even though structured prompting did not produce the best recommendation-quality scores. This is the clearest demonstration in this study's results of why a single-metric evaluation would be inadequate for this task: a marker evaluating only recommendation overlap would conclude zero-shot is the best strategy outright, while a marker evaluating only factual reliability would reach the opposite conclusion for structured prompting. The four-criteria framework adopted in Section 3.4.3, motivated directly by the gap identified in the literature review (Section 2.6), is what makes this trade-off visible.

5.1.4 RQ4: Consistency

[NOTE TO SELF/SUPERVISOR: complete this subsection once `results/consistency_scores.csv` is populated from the completed local consistency run — discuss mean pairwise Jaccard similarity by strategy, and relate to the RQ2 finding above (does the strategy with the best recommendation quality also produce the most stable recommendations, or is there a stability/quality trade-off analogous to the quality/factuality trade-off found for RQ3?).]

5.2 Relation to the Literature

The finding that prompting strategy has a large, statistically significant effect on this task (Section 4.2) is consistent with the general premise of the prompt-engineering literature (Schulhoff et al., 2024) that prompt design is a first-class variable rather than an implementation detail. However, the specific direction of the effect — zero-shot outperforming few-shot — diverges from the more common finding in general NLP benchmarks that few-shot prompting improves performance (Section 2.2), reinforcing this dissertation's positioning (Section 2.6) that recommendation tasks with a real, checkable product catalogue behave differently from the classification and QA benchmarks that dominate the existing zero-shot/few-shot comparison literature. The factual-correctness results also extend the hallucination concerns raised generally in the LLM-recommendation survey literature (Zhao et al., 2024; Wu et al., 2024) with a concrete, measured error rate specific to this domain, rather than a general acknowledgement that hallucination is possible.

5.3 Limitations

Synthetic data. All 300 user profiles are synthetic, calibrated against aggregate ONS expenditure statistics rather than drawn from real individual spending records (Section 3.2.2), because no such records are legally available for this kind of research (UK GDPR, Financial Services and Markets Act 2000). ONS calibration anchors the plausibility of the aggregate distributions used but cannot capture the irregular, bursty, or life-event-driven spending patterns a real consumer would exhibit, so findings about which prompting strategy performs best may not transfer perfectly to real users with messier spending histories.

Single model. All recommendations were generated by one model (Llama 3.1, 8B parameters). The explanation for the zero-shot finding proposed in Section 5.1 — that limited attention capacity at this model scale disadvantages longer prompts — is a testable hypothesis, not a confirmed result, because this study did not repeat the experiment with a larger model to check whether the effect persists, diminishes, or reverses at greater scale.

Baseline as reference point, not ground truth. As discussed in Section 5.1, the cosine-similarity baseline (Section 3.3) is itself a simplified, eight-dimensional approximation of "good fit," not an independently validated gold standard. Overlap with the baseline measures agreement with one particular reasonable method, not objective recommendation quality in an absolute sense.

LLM-as-judge for explanation quality. Using a second LLM to score explanation quality (Section 3.4.3) is grounded in the literature (Zheng et al., 2023) and specifically designed to reduce self-enhancement bias by using a distinct, larger judge model, but it still substitutes model judgement for the human judgement originally specified in the project's methodology, and inherits other documented LLM-judge biases (position and verbosity bias) that a human rater would not share.

Deterministic factual-correctness scope. The factual-correctness checker (Section 3.4.3, `src/factual_correctness.py`) can only verify claims that are specific, numeric, or exactly named against the card catalogue; it cannot assess vague or qualitative claims (e.g. "this card offers good value"). The reported error rates (Section 4.4) are therefore a lower bound on the true rate of factual inaccuracy in the responses, not a complete audit of every claim made.

Consistency sampling scope. The consistency evaluation (Section 3.4.4) uses a stratified 24-profile subsample rather than all 300 profiles, for the practical rate-limit reasons given there; findings about stability generalise to the sampled archetypes but were not verified across every individual profile.

Infrastructure constraints on methodology. The free Groq API tier's rate and daily-request limits (Section 3.4.1) meant that both the primary 900-call experiment and the subsequent zero-shot/few-shot correction rerun (Section 3.4.2) each took multiple days of wall-clock time to complete, constraining how many additional experimental conditions (e.g. additional models, additional prompting strategies, a full-N consistency check) could practically be included within the scope of this dissertation.

Chapter 6

Recommendations and Conclusions

6.1 Summary of Contributions

This dissertation set out to evaluate whether a free-tier, open-weight LLM (Llama 3.1, 8B parameters, via the Groq API) can generate personalised UK credit-card recommendations competitive with a conventional content-based baseline, and how prompting strategy affects the quality, factual reliability, explanation quality, and consistency of its output. Against a fixed catalogue of 20 real UK credit cards and 300 ONS-calibrated synthetic user profiles spanning eight consumer archetypes, three prompting strategies (zero-shot, structured, few-shot) were evaluated across 900 LLM inference calls, scored against a content-based cosine-similarity baseline and a four-criteria evaluation framework (recommendation quality, factual correctness, explanation quality, consistency).

The central empirical finding is that prompting strategy has a large, statistically significant effect on recommendation quality in this domain ($H = 60.70$, $p = 6.6 \times 10^{-14}$), but the effect runs in the opposite direction to what the general prompt-engineering literature would predict: zero-shot prompting outperformed both structured and few-shot prompting on recommendation-baseline overlap and top-1 accuracy. At the same time, structured prompting produced the most factually reliable responses (73.7% entirely error-free, against 55.0% for zero-shot and only 24.0% for few-shot), and few-shot prompting — despite generating the highest volume of specific factual claims — was the least reliable strategy on both recommendation quality and factual correctness. No single strategy dominates across every criterion, which is the core methodological argument this dissertation makes: a recommendation task evaluated on a single accuracy-like metric would have reached a materially incomplete, and potentially misleading, conclusion about which prompting strategy is "best."

6.2 Answering the Research Questions

RQ1 LLM-generated recommendations showed substantial but partial overlap with the content-based baseline (mean overlap 0.517–0.687 depending on strategy), confirming an LLM can approximate a purpose-built recommender without task-specific training, while not fully replicating its output.

RQ2 Prompting strategy produces a statistically significant difference in recommendation quality (Kruskal-Wallis $p < 0.0001$), with zero-shot significantly outperforming both other strategies (paired Wilcoxon $p < 0.0001$ in both comparisons) and structured vs. few-shot not reaching significance ($p = 0.0588$).

RQ3 Factual reliability of stated reasoning varies substantially and independently of recommendation quality: structured prompting is most reliable (73.7% error-free), few-shot least (24.0% error-free), demonstrating that recommendation accuracy and explanation trustworthiness are separate properties that must be evaluated separately.

RQ4 [NOTE TO SELF/SUPERVISOR: finalise once `results/consistency_scores.csv` is populated.]

6.3 Recommendations

For practitioners considering an LLM-based recommender for a similarly bounded, factually-checkable product-recommendation task: this study's results suggest that (i) prompting strategy should be treated as an empirical question to test against the specific task and model, not assumed from general-purpose benchmarks, since the direction of effect found here (zero-shot best) is the reverse of the common expectation; (ii) recommendation-quality metrics and factual-reliability metrics should be evaluated and reported separately, since a strategy that looks best on one can be worst on the other, as shown starkly by few-shot prompting in this study; and (iii) if factual reliability of the stated reasoning matters for a given application (e.g. a live consumer-facing recommender, where a factually wrong claim about a fee or reward rate is a real-world liability), structured output prompting is worth the trade-off in raw recommendation-overlap performance found here.

For future work building directly on this dissertation: the model-scale hypothesis proposed in Section 5.1.2 (that limited attention capacity at 8B parameters specifically disadvantages longer structured/few-shot prompts) could be tested directly by repeating this exact experimental design with a larger model (e.g. a 70B-parameter model, also available on Groq's free tier) to see whether the zero-shot advantage persists, narrows, or reverses at scale. The consistency evaluation could be extended from the stratified 24-profile subsample used here to the full 300-profile set, infrastructure permitting. Finally, the deterministic factual-correctness checker built for this study (`src/factual_correctness.py`) is itself a reusable contribution: it could be extended to a wider card catalogue or a different financial-product domain (e.g. savings accounts, insurance) with comparatively little modification, since its core method — extracting named/numeric claims via pattern matching and checking them against a structured ground-truth table — is domain-agnostic.

6.4 Concluding Remarks

This dissertation demonstrates that a free, open-weight LLM can perform a real, factually-grounded personalised recommendation task at a level worth taking seriously as an alternative to conventional content-based filtering, while also demonstrating, empirically and specifically for this task, why prompting-strategy choice cannot be treated as a minor implementation detail: it is the single largest source of variation in output quality found in this study, and it affects different quality dimensions — accuracy against a baseline, and factual reliability of stated reasoning — in different, sometimes opposing, directions.

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Appendix A

Repository and Code

The full source code, data, prompts, and generated results for this dissertation are available at:

`https://github.com/hindvratsingh-ux/dissertation-creditcard-llm`

The repository includes: `src/build_profiles.py` (synthetic profile generation), `src/baseline.p` (cosine-similarity baseline), `src/llm_eval.py` (the 900-call LLM experiment), `src/score_outputs.p` and `src/analyze_results.py` (recommendation-quality scoring and statistical testing), `src/factual_correctness.py` (deterministic factual-claim checking), `src/llm_judge.py` (LLM-as-judge explanation-quality scoring), `src/consistency_check.py` (repeated-run stability scoring), and `src/generate_plots.py` (figures). `docs/status-and-gap-analysis.md` in the repository documents the full development history and methodological corrections made over the course of the project.

Appendix B

Prompt Templates

B.1 Zero-Shot Prompt

```
You are a UK credit card recommendation expert.

Below is a user’s spending profile as a JSON object, followed by
the full credit card catalogue.

Your task: recommend exactly the top 3 credit cards that best match
this user’s spending habits, goals, annual fee preference, and
reward type preference.

Rules:
- Only recommend cards that exist in the catalogue below. Do NOT
  invent card IDs.
- Return exactly 3 lines, one per recommended card, ranked best
  first.
- Each line MUST use this exact format:
CC001 - <one short sentence explaining why this card fits this
user, citing the specific attribute (fee, reward rate, or offer)
that justifies it>
- Do not include any text before the first line or after the
  third line. No headings, no extra commentary.

User Profile:
{profile}

Credit Card Catalogue:
```

```
{cards_csv}
```

B.2 Few-Shot Prompt (worked examples only; rules and format identical to zero-shot above)

```
--- EXAMPLE 1 ---
User Profile:
{"profile_id": "EX001", "profile_type": "student_low_spend",
 "preferred_reward_type": "None", "annual_fee_preference": "low",
 "goal": "build_credit", ...}
Recommendation:
CC004 - No annual fee and no rewards complexity, designed
specifically for building credit history.
CC005 - No annual fee, simple entry-level card suited to a low,
irregular spender.
CC011 - Student-focused card with no fee, matching the low-income/
no-vehicle profile.

--- EXAMPLE 2 ---
User Profile:
{"profile_id": "EX002", "profile_type": "frequent_traveller",
 "preferred_reward_type": "Points", "annual_fee_preference": "high",
 "goal": "travel_rewards", ...}
Recommendation:
CC008 - Highest travel reward rate (1 point per £1 on travel) and
points transfer to airlines, matching the high travel spend and
points preference.
CC007 - Avios earned on BA spend directly rewards frequent flying,
aligned with travel_rewards goal.
CC013 - Travel-focused points card with free travel insurance,
complementing heavy travel spend.
```

B.3 Structured Prompt

The structured condition uses the same rules as zero-shot above, but specifies a JSON output schema with fields for card ID, card name, a one-sentence reason, and any relevant warnings, for each of the three ranked recommendations. See `prompts/structured.txt` in the repository for the full text.

B.4 LLM-as-Judge Prompt

See `prompts/judge_explanation_quality.txt` in the repository for the full explanation-quality scoring rubric given to the judge model (Section 3.4.3).

Appendix C

Credit Card Catalogue (Extract)

Table C.1 shows the first five entries of the 20-card catalogue used throughout this study (full catalogue in `data/cards.csv`).

Table C.1: Credit card catalogue extract

ID	Card Name	Issuer	Annual Fee (£)	Reward Type
CC001	American Express Platinum Cashback Everyday	American Express	0	Cashback
CC002	American Express Platinum Cashback	American Express	25	Cashback

Appendix D

CSV Schema Reference

Full column-level documentation for `data/cards.csv`, `data/profiles.csv`, `data/ground_truth`
and `data/llm_results.csv` is maintained in `docs/schema-notes.md` in the repository, and is not reproduced in full here for brevity.